

Molecular Sub-structural Pattern Discovery using Transformer-CNN-OMBO Algorithm for Drug-target interaction (DTI) Prediction

K. Nandhini Research Scholar

Department of Computer Science,

Vels Institute of Science, Technology & Advanced Studies

Pallavaram, Chennai, Tami l Nadu, India

kgnandhugopi@gmail.com

Dr. P. Mayilvahanan Professor & Director,

Department of Computer Science,

Vels Institute of Science, Technology & Advanced Studies

Pallavaram, Chennai, Tami l Nadu, India

mayil.scs@velsuniv.ac.in

Abstract— Predicting drug-target interactions (DTIs) is critical in drug discovery and repositioning because it identifies the reaction between a chemical molecule and a target protein. Currently, researchers use supervised learning methods based on machine learning (ML) to handle the prediction issue. Predicting drug-target interactions is hindered by an imbalanced dataset. Substructures of DTI were not included in existing molecular representation learning algorithms, which resulted in a lack of precision and difficulty in articulating. Transformer-CNN-OMBO technique for DTI interaction prediction is proposed in this study. It is first necessary to transform input drug and protein datasets to explicit sub-structures using an Improved Frequent Subsequence Mining (IFSM) technique. Then the sub-structures are sent into a transformer binding phase, which generates a contextual binding for each structure. The neighbourhood interaction was extracted by adding an optimal Convolution Neural Network (CNN) layer to these interactions and predicting the resulting interactions. During the training procedure, the Opposition based Monarch Butterfly Optimization (OMBO) algorithm is used to optimise the hyper parameters of CNN layers. Using the BindngDB dataset, the suggested technique was tested with Transformer+CNN and MPNN+CNN models. The recommended Transformer-CNN-OMBO technique achieves improved accuracy, sensitivity, specificity metrics.

Keywords—Drug-Target Interactions (DTI), Molecular Sub- structures, Improved Frequent Subsequence Mining (IFSM), Convolution Neural Network (CNN), Opposition based Monarch Butterfly Optimization (OMBO), Transformer

I. INTRODUCTION

Drug-target interactions show that the medication is related to the place where behaviour is changed by the drug. When ingested or absorbed, drugs or necessary treatments may contain a chemical compound that can affect the human body in a material way. Any region of the organism that is connected to the drug's physiological effects is a target. Detection of new drug compounds for biological purposes depends on predictions of drug- target interactions. Protein coupled receptors enzymes, and ion channels are the

most likely to be targeted drug targets today. [1]. Predicting DTIs is crucial in discovering a drug and repositioning because it identifies the response between a chemical molecule and a target. Experiments used chemical molecule and a target. Experiments used to identify DTI are expensive and take a long time before effective medications are identified [2].

Small molecule substances in the body may be targeted by biological macromolecules, which can attach to them and cause particular physiological or pharmacological effects. They are used for a variety of purposes, including disease prevention and treatment. Ion channels, enzymes, receptors, and other molecules are the most prevalent targets. Identifying new targets and relocating existing ones are critical steps in the development of new drugs. Drug discovery and drug repositioning are two separate processes that are used to identify new medications and new targets for existing drugs. There is just a small portion of the dataset's interaction relationships that have been verified so far.

Pharmacology and therapeutic medication design heavily rely on DTI prediction. There are several approaches to address this issue. The first step is to identify novel targets for an already existing medicinal compound. Secondly, if you have a certain protein in mind, you may search the library for prospective medicines. Alternatively, a pair of drugs and a target protein may be evaluated to see whether there is an interaction.

Currently, researchers use supervised learning approaches based on machine learning (ML) to tackle the prediction issue. Predicting drugs is complicated by an unbalanced dataset. There are few confirmed interactions between drug targets, therefore most techniques regard the unfamiliar interactions as negative values and hence they dominate the positive ones. Another problem is how the drug-target pair is stored in the input dataset. Predicting DTIs using ML techniques commonly uses molecular traces of medicines and protein sequence data [4].

Deep Learning (DL) and other ML techniques have been applied in various medical applications such as automatic segmentation of CT images etc [12] [13]. DTIs prediction is

a biological application of DL-based approach, which can learn architectures of data with many abstraction categories. DL-based approaches for large-scale DTI predictions, such as DeepDTI and GraphDTA, have recently been developed. [5][6].

A. Objectives

The main objectives of this research work are

- To design deep learning based sub- structural pattern discovering algorithm for more accurate DTI prediction.
- To determine the semantic relationships among the substructures extracted from huge unlabelled data.

II. LITERATURE SURVEY

There is a new classifier for predicting drug-target interactions, FRnet-DTI, developed by Farshid Rayhan et al [4]. The FRnet-Encode and FRnet- Predict CNNs are being considered. One model is used to manipulate features, while the other is used to classify them. There are 4096 characteristics for each dataset, and the second approach, FRnet- Predict, uses those features to determine how likely it is that two datasets will interact. Four gold standard datasets used extensively by other researchers have been used to evaluate their strategy.

Analysis of the three main reasons why machine learning has become so popular has been completed by Lei Xu et al [7]. They summarise the explicit applications of these methods in each step of constructing ML models, analyse the shortcomings and discuss several aspects that can be further explored.

The validity of the suggested criteria has been examined by Stephen Scott MacKinnon et al [8]. They pointed out that the current literature had several flaws, making it impossible to assess the practical efficacy of some of the strategies presented only on the basis of their writing. There is a hope that this work may assist to fix this problem and raise awareness about many forms of bias that may fall into frequently used cross- validation techniques.

The data needed for DTI prediction has been described by Maryam Bagherian et al [9], followed by a complete catalogue of ML algorithms and datasets that have been suggested and used to predict DTIs. Additionally, the merits and drawbacks of each set of procedures are briefly reviewed. Finally, the difficulties that might arise when attempting to predict DTI using machine learning algorithms are discussed, and some key future research avenues are highlighted as a result.

A. Limitations of Existing Works

Existing molecular representation learning techniques did not consider the sub-structures of DTI, resulting in less accuracy and difficult to express

Existing technique mostly concentrate on less number of labelled thereby omitting the value of huge unlabelled data.

III. PROPOSED METHODOLOGY

A. Overview

In this paper, we propose to design Transformer CNN–OMBO algorithm for DTI prediction, the architecture of which is shown in Figure 1.

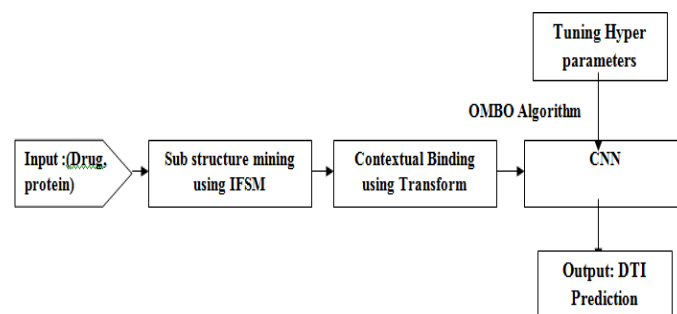


Figure 1 Architecture Transformer-CNN–OMBO

For a pair of drug and protein data, an IFSM algorithm converts the original drug-target data to a sequence of specific substructures (S_D, S_T). Then these sub-structures are entered into a transformer binding phase to obtain a contextual binding for each sub-structure. The interaction prediction phase contains two layers: (i) an interaction layer which models the interactions of each sub-sequence in protein and drug (ii) an optimized CNN layer (OCNN) over these interaction to extract the neighborhood interaction. Hence in this paper, the OMBO algorithm is applied for tuning the hyper parameters of CNN layers during training process.

B. Improved Frequent Sub-sequence Mining (IFSM)

Here, we provide an IFSM algorithm for identifying repeating subsequences in drug and protein databases. IFSM breaks down each protein or drug sequence into mini subsequences and separate atoms/amino acids notations in a hierarchical manner.

Initialization of a terminology set V of distinct amino acid tokens or SMILES strings is the first step in a smart contract for the complete drug/protein corpus. W is the name given to the tokenized set.

Identification of repeating sequence: W is then scanned, and the most often occurring sequence of tokens is identified as follows: (A, B). The new token (AB) is then added to vocabulary set V by FCS, and each (A, B) in the tokenized set W is modified to reflect the new token (AB). The scanning, identifying and updating methods are repeated if there are no frequent tokens over an upper bound h or if the size of V exceeds a built-in maximum value l . In this procedure, the most often occurring sub-sequences are combined into a single token, while the less frequently occurring sub-sequences are broken down into a series of smaller tokens.

Finally, for each pair, IFSM returns a sequence $C =$

$\{C_1, \dots, C_k\}$ of substructural pairs with size k , where $C_i \in V$. Thus, this algorithm transforms the input pair into a sequence of known substructures (S_D, S_T).

IFSM is capable of capturing biomedical semantics that are basic and meaningful. An important part of drugs and proteins is found in the generated sub-structures. Given a variety of dataset features, the FCS algorithm is able to identify a comparable collection of basic biochemical substructures regardless of the kind of organisms included in the protein dataset or the probability of the medicine to be successful.

C. Transformer Binding Phase

The outputs of IFSM algorithm are entered into a transformer binding phase to get a contextual binding for each chemical sub-structure. It first initializes a Learnable Lookup Dictionary (LLD) for each substructure and then enhance the binding with the contextual substructural details using transformer encoders.

The transformer encoder is a DL architecture that enforces the self-governance technique to generate contextual binding for learning the molecule architecture. The self-governance technique alters each sub-structure binding by exploring from all other sub-structures of the similar molecule.

For each input (D, T) pair, the associated sequence of sub-structures (S_D, S_T) are transformed into the matrices (M_D, M_T).

Let Z_D and Z_T are the size of the sub-structures of drug and target.

Let L_D and L_T are the lengths of the substructure sequences of (D, T).

The content binding $Cont_D$ and $Cont_T$ for each drug target pair is generated using LLD as

$$\begin{aligned} Cont_{iD} &= MCont_D . M_{iD} \\ Cont_{jT} &= MCont_T . M_{jT} \end{aligned} \quad (1)$$

Where $MCont_D$ and $MCont_T$ are matrices of LLD and M_{iD} and M_{jT} represent the vector of i^{th} sub-structure of drug and j^{th} sub-structure of target.

The final binding of $BCont_D$ and $BCont_T$ are generated as

$$\begin{aligned} BCont_D &= Cont_{iD} + Pos_{iD} \\ BCont_T &= Cont_{jT} + Pos_{jT} \end{aligned} \quad (2)$$

Where Pos_{iD} and Pos_{jT} are positional bindings determined from LLD.

D. Interaction Prediction using CNN.

A CNN is a network with a limited level of complexity and with multiple layers. It has a compound technical model for analyzing the data in an error mode. CNN with many

layers usually integrates the properties removal and combining task into a learning entity.

In common, A CNN contains the following layers:

Convolutional Layer (CL): It uses n number of filters to the feature map. After the intricacy, A Relu stimulation function is used to include non-linearity to the network.

Max Pooling Layer (MPL): Max Pooling is the traditional method to divide the feature maps into sub regions (with a 2×2 size) and retains only the extreme values.

Fully Connected layer (FCL): The whole neurons from the preceding layers are associated to the subsequent layers. The CNN will categorize the labels based on the aspects from the CLs and abridged with the PL.

In the traditional CNN training process, the factors of convolutional kernels and hidden neurons in the fully connected (FCL) layer are adjusted. Mostly, Stochastic Gradient Descent (SGD) method is applied in CNN for adjusting the convolution layer (CL) and FCL factors. But SGD contains many hyper parameters that may affect the network's performance. Hyper parameters are responsible for determining the accuracy and convergence of the CNN [10].

Table 1 illustrates the details of the CNN architecture used for interaction prediction.

Table 1 Detailed Parameters of CNN model

Layer	Weight Filter	Size	Stride
Input	224x224x3		
CL1	64:3 x 3	1x1	
MPL1	2 x 2		2x2
CL2	64:3 x 3	1x1	
MPL 2	2 x 2		2x2
CL3	32:3 x 3	1x1	
MPL3	2 x 2		2x2
CL4	16:3 x 3	1x1	
MPL4	2 x 2		2x2
CL5	8:3 x 3	1x1	
FC	Output size =3		
	Output SoftMax		

E. Optimizing CNN using OMBO Algorithm

In this section, the following hyper parameters of CNN are optimized using OMBO algorithm to give most accurate results

- Learning rate (LR): It controls the speed of SGD algorithm
- Number of epochs (E): It computes the number

of updations of the network parameters by the learning algorithm

- Regularization coefficient (RC): It avoids the overfitting problem of the network.

A fitness function is derived in terms of these parameters as

$$\text{Fitness}() = (\alpha.LR + \beta.E + \gamma.RC) \quad (3)$$

Where α , β , γ are weighting constants ranging from (0,1).

It has been proved that OMBO algorithm is an efficient solution to solve different types of optimization problems. It can minimize the sorting steps and improve the computing efficiency. This research work develops OMBO algorithm based on relocation operator and butterfly regulating operator.

The OMBO algorithm for automatic tuning of hyperparameters in SGD training is explained below. By adaptively, varying the learning rate of CNN layers, the speed of SGD can be controlled by the OMBO algorithm.

a. Initializations

Initial solutions are produced based on weights and the size of the solution is computed

Let SP1 and SP2 be the sub populations of the initial population P

The following equation integrates the opposition based solution with the initial solution as

$$O_i = x + y - I_i \quad (4)$$

where $I_i \in I_1, I_2, I_3, \dots, NP$ are initial results which are arbitrarily produced.

b. Butterfly Migration Operator (BMO)

For each i in SP1, the position from t^{th} and to $(t + 1)^{\text{th}}$ generation is updated by Eq. (2):

$$x_{i,k}^{t+1} \leftarrow \begin{cases} x_{r1,k}^t & \text{if } r \leq p \\ x_{r2,k}^t & \text{else} \end{cases} \quad (5)$$

Where $r1, r2$ are randomly picked from SP1 and SP2, respectively, p is the ratio of butterflies in SP1.

r is computed as

$$r = (\text{rand}) * mp$$

where mp denotes the displacement period.

c. Butterfly Adjustment Operator (BAO)

For each i in SP2, the position from generation t to $t + 1$ is given as follows:

$$x_{i,k}^{t+1} \leftarrow \begin{cases} x_{best,k}^t & \text{if } rand \leq p \\ x_{r3,k}^t & \text{if } rand > p \wedge rand \leq BAR \\ x_{i,k}^t + \alpha X(d_{xk} - 0.5) & \text{if } rand > p \wedge rand > BAR \end{cases} \quad (6)$$

Where $x_{best,k}^t$ is the k^{th} term of the present global best solution in P at t , $r3$ is randomly picked from SP2, BAR is butterfly adjustment rate.

The parameters α and dx are weight constants computed by the following equations

$$\alpha = S_{\max} / t^2 \quad (7)$$

$$dx = \text{Levy}(x_i^t) \quad (8)$$

Where $S_{\max}$ is max walk step, Levy indicates the Levy flight movement process of butterflies.

d. Opposition based learning

Opposition-based learning (OBL) is applied in a variety of optimization strategies to handle a variety of practical problems. The opposing solution may be assessed simultaneously since picking a solution contains some randomness in the optimization process. In most cases, OBL may amplify the chances of determining the best solution and speeding up convergence. It's a space transformation approach coupled with a generalised opposition-based learning technique. A generalised OBL may have a lot of opportunities to locate the candidate solutions that are closer to the global best. The new opposition point X is denoted as below:

$$H'(x_i) = \{x_i' | x_i' = r^* (a_i + b_i) - x_i, x_i \in R^N, x_i \in [a_i, b_i], i \in \{1, 2, \dots, n\}\} \quad (9)$$

where r is randomly picked.

Since, OBL amplifies the chances of determining the best solution and speeds up the convergence process, the OMBO optimization significantly reduces the computational complexity and increases the accuracy of CNN prediction.

e. OMBO Algorithm

Begin

1. Sort all items in the descending order by p_i/w_i , ($1 < i < n$) and the indices are stored in an array $H[1...n]$
2. Initialization.
 - 2.1 Generate NP monarch butterflies $\{<X_1, Y_1>, <X_2, Y_2>, \dots, <X_{NP}, Y_{NP}\}$, randomly.
 - 2.2 Compute the fitness of each solution, $f(Y_i)$, $1 < i < NP$.
 - 2.3 Divide the P into SP1 and SP2.
 - 2.4. Let $\max_iter = \text{MaxGen}$ and iter no $g=1$.
 - 2.5. Set the generating interval of recombining two subpopulations RG.
 - 2.5 Set the $Ga=0.2$.
- 3 While ($g < \max Gen$ or stopping criterion is not satisfied)
 - 3.1 Evaluate the fitness of the swarm and store the $<X_{gbest}, Y_{gbest}>$
 - 3.2 Update SP1 with BMO.
 - 3.3 Update SP2 with BAO.
 - 3.4 Update the worst $Ga * NP$ individual with Gaussian perturbation.
 - 3.5 Utilize greedy modification operator (GMO) to correct the infeasible solutions.
 - 3.6 Utilize greedy optimization operator (GOO) to improve the feasible solutions.
 - 3.7 Store the best solutions.
 - 3.8 Determine the current best ($Y_{gbest}, f(Y_{gbest})$).
 - 3.9 $g=g+1$
 - 3.10. (if $g \geq \text{MaxGen}/2$)
 - 3.10.1 Perform OBL on the latter half of the individuals.
 - 3.10.2. Repair the infeasible solutions using GMO
 - 3.10.3 Improve the feasible solutions using GOO
 - end if
 - 3.11 if $\text{Mod}(g, RG)=0$
 - 3.11.1 Recombine Subpopulation_1 and subpopulation_2 into one.
 - 3.11.2 Split the population into two subpopulations based on fitness value
 - End if
4. end while
5. Return the best results
6. End

IV. EXPERIMENTAL RESULTS

A. Specification of Dataset

We have implemented the proposed Molecular Sub-structural Pattern Discovery (MSPD) algorithm in DeepPurpose toolkit. DeepPurpose is a pytorch-based deep learning framework designed to provide a powerful toolkit for DTI prediction.

The BindingDB dataset from <https://www.bindingdb.org/bind/index.jsp> has been used in the experiments. The input to the model is a drug-target (DT) pair, in which drug uses SMILES string (d) and the target (t) uses the Amino acid sequence. The output (y) is the score which is

the binding action of the (d,t) pair. $Y=0$ or 1 representing the interaction output indicating the binding value.

The format of the training dataset is shown below:

```
Drug-1:
CC1=C2C=C(C=CC2=NN1)C3=CC(=CN=C3)OCC(CC4=CC=CC=C4)N
Target-1:
MKKFFDSRREQGSGSLGSGSSGGGGSTSGLGSGYIGRVFGI
GRQQVTVDVLAEGGFAIVFLVRTSNGMKCALKRMFVNNEH
DLQVCKREIQIMRDLSGHKNIVGYIDSSINNVS S G D V W E V L
ILMDFCRGGQV V N L M N Q R L Q T G F T E N E V L Q I F C D T C E A V A R
LHQCCTPIIHRDLKVENILLHDRGHYVLCDFGSATNKFQNP
QTEGVNAVEDEIKKYTTLSYRAPEMVNLYSGKIIITTKADIW
ALGCLLYKLCYFTLPFGESQVAICDGNFTIPDNSRYSQDMH
CLIRYMLEPDPDKRPDIYQVSYSFKLLKKECPIPNVQNSP
IPAKLPEPVKASEAAAKKTQPKARLTDPIPTTETSIAPRQR
PKAGQTQPNPGILPIQPALTPRKRATVQPPPQAAGSSNQPG
LLASVPQPKPQAPPSQPLPQTQAKQPQAPPTPQQTPSTQAA
GLPAQAQATPQHQQQLFLKQQQQQQQPPPAQQQPAGTFYQQ
QQAQTQQFQAVHPATQKPAIAQFPVVSQGGSQQLMQNFYQ
QQQQQQQQQQQQQLATALHQQLMTQQAALQQKPTMAAGQQ
PQPQAAAAPQAPAPAEPAIQAPVRQQPKVQTTTPPAVQGGK
VGSLTPSSPKTQRAGHRRILSDVTHSAVFVGPASKSTQLL
QAAAAEASLNKSKSATTTTPSGSPRTSQQNVYNPSEGSTWNP
FDDDNFSKLTAEELLNKDFAKLGEGKHPEKLGGSAESLIPG
FQSTQGDAFATTSFSAGTAEKRKGQTVD S G L P L L S V S D P F
IPLQVPDAPEKLEGLKSPDTSLLLPDLLPMTDPFGSTSDA
VIEKADVAVESLIPGLEPPVQRLPSQTESVTSNRTDSL TG
EDSLLDCSLLSNPTTDLLEEFAPTAISAPVHKAAEDSNLIS
GFDVPEGSDKVAEDEFDPIPVLTITKNPQGGHSRNSSGSSSES
SLPNLARSLLLVDQLIDL
```

Figure 2 Training data set

B. Results of Encoding

The encoders used for drug and target are MPNN and CNN, respectively. It encodes the data into the specified format, supplies the training and testing split fractions. The output of the drug and target encoding process is shown below:

```
in total: 1 drug-target pairs
encoding drug...
unique drugs: 1
drug encoding finished...
encoding protein...
unique target sequence: 1
protein encoding finished...
splitting dataset...
do not do train/test split on the data for already splitted data
predicting...
The predicted score is [5.243870735168457]
```

Figure 3 Output of Encoding process


```

Training at Epoch 96 iteration 0 with Loss 0.134930276
Validation at Epoch 96 , MSE: 0.3258646538582363 , Pearson Correlation: 0.7833035914012831 with p-value: 0.0 , Concordance Index: 0.8716263817735894
Training at Epoch 97 iteration 0 with Loss 0.14196697
Validation at Epoch 97 , MSE: 0.2894430110949083 , Pearson Correlation: 0.7894924155305123 with p-value: 0.0 , Concordance Index: 0.864822823670595
Training at Epoch 98 iteration 0 with Loss 0.0763197
Validation at Epoch 98 , MSE: 0.2824313373215666 , Pearson Correlation: 0.7947889487940375 with p-value: 0.0 , Concordance Index: 0.8668753821178348
Training at Epoch 99 iteration 0 with Loss 0.0673421
Validation at Epoch 99 , MSE: 0.3075603177427848 , Pearson Correlation: 0.7977542563703856 with p-value: 0.0 , Concordance Index: 0.8696652822577812
Training at Epoch 100 iteration 0 with Loss 0.11071129
Validation at Epoch 100 , MSE: 0.2819463625926039 , Pearson Correlation: 0.7962812176681273 with p-value: 0.0 , Concordance Index: 0.860235339812648
--- Go for Testing ---
Testing MSE: 0.2565965372661101 , Pearson Correlation: 0.8231646722825967 with p-value: 0.0 , Concordance Index: 0.8681803072880361
--- Training Finished ---

```

Figure 4 Output of training process

The accuracy of the training process is validated against 600-00 epochs and the corresponding loss values are measured and plotted in the following figure.

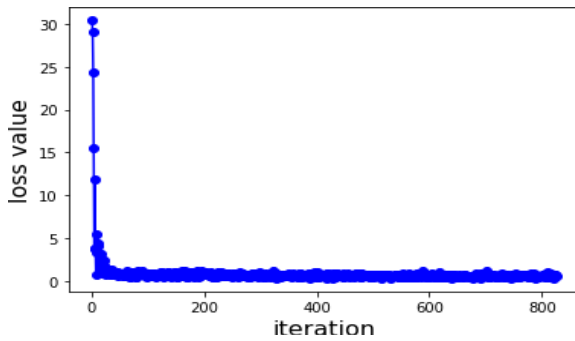


Figure 5 Validation graph for training

C. Results of Prediction using CNN-OMBO

For the sample data from BindingDB dataset, a list of DT pairs that have high binding desires are generated as follows.

Rank	Drug Name	Target Name	Binding Score
1	Drug 80	Target 80	0.58
2	Drug 86	Target 86	0.85
3	Drug 4	Target 4	0.89
4	Drug 89	Target 89	1.62
5	Drug 44	Target 44	1.67
6	Drug 58	Target 58	1.87
7	Drug 35	Target 35	2.74
8	Drug 43	Target 43	2.98
9	Drug 23	Target 23	3.95
10	Drug 25	Target 25	4.55
11	Drug 15	Target 15	4.86
12	Drug 97	Target 97	5.03
13	Drug 24	Target 24	6.31
14	Drug 47	Target 47	6.53
15	Drug 48	Target 48	11.22
16	Drug 50	Target 50	11.35
17	Drug 92	Target 92	16.71
18	Drug 66	Target 66	17.49
19	Drug 26	Target 26	18.03
20	Drug 87	Target 87	18.64
21	Drug 93	Target 93	19.38
22	Drug 20	Target 20	23.28
23	Drug 49	Target 49	27.10
24	Drug 30	Target 30	28.13
25	Drug 33	Target 33	33.48

Figure 7 Results of Virtual screening for BindingDB dataset

D. Comparisons with Other algorithms

In this section, the prediction results of Transformer-CNN-OMBO algorithm is compared with the traditional Transformer+CNN and MPNN+CNN in which Message Passing Neural Network (MPNN) is applied for encoding.

The performance is evaluated in terms of the metrics accuracy, sensitivity, specificity, precision, recall and F1-Score which is specified by the following formula.

Table 2 Performance Comparison with other algorithms

Metrics	Transformer+CNN+OMBO	Transformer+CNN	MPNN+CNN
Recall	0.90	0.89	0.86
Precision	0.43	0.41	0.39
Accuracy	0.81	0.78	0.76
Sensitivity	0.80	0.77	0.75
Specificity	0.90	0.87	0.84
F1 score	0.59	0.57	0.54

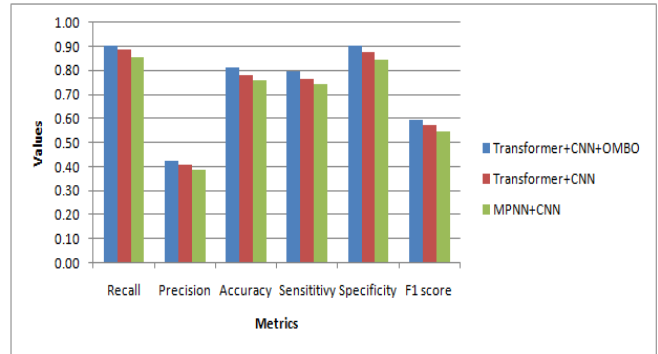


Figure 8 Performance comparison of Transformer+CNN+OMBO

V. CONCLUSION

In this paper, we propose to design Transformer CNN-OMBO algorithm for DTI interaction prediction. Initially, for the input dataset, an IFSM algorithm is used to convert the DTpair to a sequence of well known sub-structures which are then entered into a transformer binding phase to determine a contextual binding for each sub-structure. The interaction prediction was done by optimized CNN layer over these interactions to extract the neighborhood interaction. The OMBO algorithm is applied for optimizing the hyper parameters of CNN layers during training process. The proposed algorithm has been applied on the BindingDB dataset and its performance was compared with Transformer+CNN and MPNN+CNN models. Results have shown that the proposed Transformer-CNN-OMBO algorithm outperforms the other models by achieving high accuracy, sensitivity, specificity, recall, precision and F1-score metrics.

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